ZEARN Math

Tennessee State Standards Alignment

For Grades K-8

About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Tennessee's Learning Standards for Mathematics.

We are committed to ensuring that our high-quality instructional materials align to Tennessee's Learning Standards for Mathematics to fully support Tennessee's students and teachers. The tables on the following pages, organized by Tennessee Learning Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
Kindergarten Standards		**Lessons**
Counting & Cardinality		
K.CC.A.1	Count to 100 by ones, fives, and tens. Count backward from 10.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 15-18
K.CC.A.2	Count forward by ones beginning from any given number within the known sequence (instead of having to begin at 1).	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 16-18
K.CC.A.3	Write numbers from 0 to 20. Represent a quantity of objects with a written number 0-20.	Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 24-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 4, Topic A, Lesson 1-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic E, Lesson 25, 27 Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 6, 9 Mission 5, Topic C, Lesson 10-13
K.CC.A.4	Recognize, describe, extend, and create patterns and explain a simple rule for a pattern using concrete materials. Analyze the structure of the repeating pattern by identifying the unit (core) of the pattern.	Mission 1, Topic H, Lesson 35 Mission 4, Topic B, Lesson 10, 12 Mission 4, Topic H, Lesson 38 Mission 5, Topic C, Lesson 11-12
K.CC.B.5	Understand the relationship between numbers and quantities; connect counting to cardinality.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic C, Lesson 7-11 Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 4, Topic H, Lesson 38
K.CC.B.6	Count to answer "how many?" questions about as many as 20 things arranged in a line, a rectangular array, a circle, or as many as 10 things in a scattered	Mission 1, Topic C, Lesson 8-11 Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
Kindergarten Standards		**Lessons**
	configuration. Given a number from 1-20, count out that many objects.	Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 3, Topic F, Lesson 23-24 Mission 5, Topic A, Lesson 1-5 Mission 5, Topic B, Lesson 6 Mission 5, Topic C, Lesson 10-14 Mission 5, Topic D, Lesson 18-19
K.CC.C.7	Identify whether the number of objects in one group is greater than, less than, or equal to the number of objects in another group.	Mission 3, Topic E, Lesson 17-19 Mission 3, Topic F, Lesson 20-24 Mission 3, Topic G, Lesson 25-28
K.CC.C.8	Compare two given numbers up to 10, when written as numerals, using the terms greater than, less than, or equal to.	Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 25-28
Operations & Algebraic Thinking		
K.OA.A.1	Represent addition and subtraction with objects, fingers, drawings, acting out situations, verbal explanations, expressions, or equations.	Mission 4, Topic C, Lesson 13-16 Mission 4, Topic D, Lesson 20-24 Mission 4, Topic F, Lesson 29-30 Mission 4, Topic G, Lesson 33 Mission 4, Topic H, Lesson 38
K.OA.A.2	Add and subtract within 10 to solve contextual problems with result/total unknown involving situations of add to, take from, and put together/take apart. Use objects, drawings, or equations to represent the problem.	Mission 4, Topic C, Lesson 16-18 Mission 4, Topic D, Lesson 19, 21 Mission 4, Topic F, Lesson 31 Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37
K.OA.A.3	Decompose numbers less than or equal to 10 into addend pairs in more than one way (e.g., 5 = 2 + 3 and 5 = 4 + 1) by using objects or drawings. Record each decomposition using a drawing or writing an equation.	Mission 1, Topic C, Lesson 9-11 Mission 1, Topic D, Lesson 14-15 Mission 4, Topic A, Lesson 2, 4-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic D, Lesson 22-24 Mission 4, Topic E, Lesson 25-28 Mission 4, Topic F, Lesson 32
K.OA.A.4	Find the number that makes 10, when added to any given number, from 1 to 9 using objects or drawings. Record the answer using a drawing or writing an equation.	Mission 4, Topic H, Lesson 39-40

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
Kindergarten Standards		**Lessons**
K.OA.A.5	Use mental strategies flexibly to develop fluency in addition and subtraction within 10.	Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37, 39 Mission 6, Topic A, Lesson 2-4 Mission 6, Topic B, Lesson 5
Numbers & Operations in Base 10		
K.NBT.A.1	Compose and decompose numbers from 11 to 19 into a group of ten ones and some more ones by using objects or drawings (e.g., 18 equals 10 + 8). Record the composition or decomposition using a drawing or by writing an equation.	Mission 5, Topic A, Lesson 2-5 Mission 5, Topic B, Lesson 6-9 Mission 5, Topic E, Lesson 20-21, 23
Measurement & Data		
K.MD.A.1	Describe the measurable attributes of an object, such as length (long/short), height (tall/short), or weight (heavy/light).	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic C, Lesson 8 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16
K.MD.A.2	Directly compare two objects with a measurable attribute in common, to describe which object has more of/less of the attribute. For example, directly compare the heights of two children and describe one child as taller/shorter.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 4-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 28 Mission 3, Topic H, Lesson 29-32
K.MD.B.3	Identify the penny, nickel, dime, and quarter based on their attributes (size and color) and recognize the value of each.	Mission 5, Topic E, Lesson 20, 23 Mission 6, Topic A, Lesson 1
K.MD.C.4	Sort a collection of objects into a given category, with 10 or fewer in each category. Compare the categories by group size.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7 Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 7
Geometry		
K.G.A.1	Describe objects in the environment using names of shapes and solids (squares, circles, triangles, rectangles, hexagons, cubes, cones, cylinders, and	Mission 2, Topic A, Lesson 5 Mission 2, Topic B, Lesson 8 Mission 2, Topic C, Lesson 10

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
Kindergarten Standards		**Lessons**
	spheres). Describe the relative positions of these objects using terms such as above, below, beside, in front of, behind, between, and next to.	
K.G.A.2	Correctly name shapes and solids (squares, circles, triangles, rectangles, hexagons, cubes, cones, cylinders, and spheres) regardless of their orientations or overall size.	Mission 2, Topic A, Lesson 2-5 Mission 2, Topic B, Lesson 7-8 Mission 2, Topic C, Lesson 9-10
K.G.A.3	Identify shapes (squares, circles, triangles, rectangles, and hexagons) as two-dimensional and solids (cubes, cones, cylinders, and spheres) as three- dimensional.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 6 Mission 2, Topic C, Lesson 9-10
K.G.B.4	Describe similarities and differences between two- and three-dimensional shapes/solids, in different sizes and orientations.	Mission 2, Topic A, Lesson 1-4 Mission 2, Topic B, Lesson 6-7 Mission 2, Topic C, Lesson 9-10 Mission 6, Topic A, Lesson 3
K.G.B.5	Model shapes/solids in the world by building or drawing them.	Mission 6, Topic A, Lesson 1-3
K.G.B.6	Compose a figure using simple shapes/solids and identify smaller shapes/solids within the figure.	Mission 6, Topic B, Lesson 5-7

1st Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
1st Grade Standards		Lessons
Operations & Algebraic Thinking		
1.OA.A.1	Add and subtract within 20 to solve contextual problems, with unknowns in all positions, involving situations of add to, take from, put together/take apart, and compare. Use objects, drawings, and equations with a symbol for the unknown number to represent the problem.	Mission 1, Topic B, Lesson 4-8 Mission 1, Topic C, Lesson 9-13 Mission 1, Topic G, Lesson 25 Mission 1, Topic H, Lesson 28-32 Mission 2, Topic A, Lesson 3-5, 7-11 Mission 2, Topic B, Lesson 12-13, 17-18, 21 Mission 2, Topic C, Lesson 23-24 Mission 2, Topic D, Lesson 27-29 Mission 3, Topic D, Lesson 10-13 Mission 4, Topic E, Lesson 19-22 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic F, Lesson 25-27
1.OA.A. 2	Add three whole numbers whose sum is within 20 to solve contextual problems using objects, drawings, and equations with a symbol for the unknown number to represent the problem.	Mission 2, Topic A, Lesson 1
1.OA.B.3	Apply properties of operations (additive identity, commutative, and associative) as strategies to add and subtract.	Mission 1, Topic E, Lesson 19-20 Mission 1, Topic F, Lesson 21-24 Mission 2, Topic A, Lesson 2, 6, 10-11
1.OA.B.4	Understand the relationship between addition and subtraction by representing subtraction as an unknown-addend problem. For example, to solve 10 – 8 = __, a student can use 8 + __ = 10.	Mission 1, Topic G, Lesson 26-27 Mission 1, Topic H, Lesson 29-30, 32 Mission 2, Topic B, Lesson 14-19, 21 Mission 2, Topic C, Lesson 22
1.OA.C.5	Add and subtract within 20 using strategies such as counting on, counting back, making 10, related known facts, and composing/decomposing numbers with an emphasis on making ten (e.g., 13 – 4 = 13 – 3 – 1 = 10 –	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-8 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic G, Lesson 26-27 Mission 2, Topic A, Lesson 5, 9

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
1st Grade Standards		**Lessons**
	1 = 9 or adding 6 + 7 by creating the known equivalent 6 + 4 + 3 = 10 + 3 = 13 OR 6 + 6 + 1 = 12 + 1).	Mission 2, Topic B, Lesson 12-17, 19, 21
1.OA.C.6	Use mental strategies flexibly and efficiently to develop fluency in addition and subtraction within 20. By the end of grade 1, know all sums and differences up to 10.	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.OA.D.7	Understand the meaning of the equal sign (e.g., 6 = 6; 5 + 2 = 4 + 3; 7 = 8 – 1). Determine if equations involving addition and subtraction are true or false.	Mission 1, Topic E, Lesson 17-19 Mission 2, Topic A, Lesson 6, 10 Mission 2, Topic B, Lesson 20 Mission 2, Topic C, Lesson 25
1.OA.D.8	Determine the unknown whole number in an addition or subtraction equation with sums/differences within 20, with the unknown in any position (e.g., 8 + � = 11, 5 = � – 3, 6 + 6 = �).	Mission 1, Topic D, Lesson 16 Mission 1, Topic H, Lesson 29-30, 32 Mission 2, Topic A, Lesson 10, 20
Numbers & Operations in Base 10		
1.NBT.A.1	Count to 120, by ones, twos, and fives starting at any multiple of that number. Count backward from 20. Read and write numbers to 120 and represent a quantity of objects with a written number.	Mission 2, Topic A, Lesson 10, 20 Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 4, 7, 9
1.NBT.A.2	Recognize, describe, extend, and create patterns when counting by ones, twos, fives, and tens and use those patterns to predict the next number in the counting sequence up to 120 through counting or building with concrete materials.	Mission 2, Topic B, Lesson 16 Mission 4, Topic D, Lesson 13 Mission 6, Topic B, Lesson 5
1.NBT.B.3	Know that the digits of a two-digit number represent groups of tens and ones (e.g., 39 can be represented as 39 ones, 2 tens and 19 ones, or 3 tens and 9 ones).	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
1st Grade Standards		**Lessons**
1.NBT.B.4	Compare two two-digit numbers based on the meanings of the digits in each place and use the symbols >, =, and < to show the relationship.	Mission 4, Topic B, Lesson 7-10 Mission 6, Topic B, Lesson 6
1.NBT.C.5	Add a two-digit number to a one-digit number and a two-digit number to a multiple of ten (within 100). Use concrete models, drawings, strategies based on place value, properties of operations, and/or the relationship between addition and subtraction to explain the reasoning used.	Mission 4, Topic C, Lesson 11-12 Mission 4, Topic D, Lesson 13-18 Mission 4, Topic F, Lesson 23-29 Mission 6, Topic C, Lesson 10-17 Mission 6, Topic D, Lesson 18-19
1.NBT.C.6	Mentally find 10 more or 10 less than a given two-digit number without having to count by ones and explain the reasoning used.	Mission 4, Topic A, Lesson 5-6 Mission 6, Topic B, Lesson 5
1.NBT.C.7	Subtract multiples of 10 from any number in the range of 10-99 using concrete models, drawings, strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 4, Topic C, Lesson 11 Mission 6, Topic C, Lesson 10
Measurement & Data		
1.MD.A.1	Order three objects by length. Compare the lengths of two objects indirectly by using a third object. For example, to compare indirectly the heights of Bill and Susan: if Bill is taller than mother and mother is taller than Susan, then Bill is taller than Susan.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9
1.MD.A.2	Measure the length of an object using non- standard units (paper clips, cubes, etc.) and express this length as a whole number of units.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-9
1.MD.B.3	Recognize a clock as a measurement tool. Tell and write time in hours and half-hours using analog and digital clocks.	Mission 3, Topic C, Lesson 8 Mission 5, Topic D, Lesson 10-13 Mission 6, Topic E, Lesson 20-24
1.MD.B.4	Count the value of a set of like coins less than one dollar using the ¢ symbol only.	Mission 3, Topic C, Lesson 8 Mission 5, Topic D, Lesson 10-13 Mission 6, Topic E, Lesson 20-24
1.MD.C.5	Organize, represent, and interpret data with up to three categories using pictographs, bar graphs, and tally charts. Ask and answer questions about the total	Mission 3, Topic D, Lesson 10-13

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
1st Grade Standards		**Lessons**
	number of data points, how many in each category, and how many more or less are in one category than in another.	
Geometry		
1.G.A.1	Distinguish between attributes that define a shape (e.g., number of sides and vertices) versus attributes that do not define the shape (e.g., color, orientation, overall size); build and draw two-dimensional shapes to possess defining attributes.	Mission 5, Topic A, Lessons 1-3
1.G.A.2	Create a composite figure and use the composite figure to make new figures by using two-dimensional shapes (rectangles, squares, hexagons, trapezoids, triangles, half-circles, and quarter-circles) or three-dimensional solids (cubes, spheres, rectangular prisms, cones, and cylinders). Partition circles and rectangles into two and four equal shares, describe the shares using the words halves, fourths, and quarters, and use the phrases half of, fourth of, and quarter of. Describe the whole as two of, or four of, the shares. Understand for these examples that partitioning into more equal shares creates smaller shares.	Mission 5, Topic B, Lessons 4-6
1.G.A.3	Create a composite figure and use the composite figure to make new figures by using two-dimensional shapes (rectangles, squares, hexagons, trapezoids, triangles, half-circles, and quarter-circles) or three-dimensional solids (cubes, spheres, rectangular prisms, cones, and cylinders). Partition circles and rectangles into two and four equal shares, describe the shares using the words halves, fourths, and quarters, and use the phrases half of, fourth of, and quarter of. Describe the whole as two of, or four of, the shares. Understand for these examples that partitioning into more equal shares creates smaller shares.	Mission 5, Topic C, Lessons 7-9 Mission 5, Topic D, Lessons 10

2nd Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
Operations & Algebraic Thinking		
2.OA.A.1	Add and subtract within 100 to solve one- and two-step contextual problems, with unknowns in all positions, involving situations of add to, take from, put together/take apart, and compare. Use objects, drawings, and equations with a symbol for the unknown number to represent the problem.	Mission 1, Topic B, Lesson 5-6 & 8 Mission 4, Topic A, Lesson 1-5 Mission 4, Topic B, Lesson 6-10 Mission 4, Topic C, Lesson 11-14, 16 Mission 4, Topic F, Lesson 31 Mission 7, Topic B, Lesson 7-8, 12-13
2.OA.B.2	Fluently add and subtract within 30 using mental strategies. By the end of 2nd grade, know all sums of two one-digit numbers and related subtraction facts.	Mission 1, Topic A, Lesson 1-2 Mission 1, Topic B, Lesson 4, 7
2.OA.C.3	Determine whether a group of objects (up to 20) has an odd or even number of members by pairing objects or counting them by 2s. Write an equation to express an even number as a sum of two equal addends.	Mission 6, Topic D, Lessons 17-19
2.OA.C.4	Use repeated addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns; write an equation to express the total as a sum of equal addends.	Mission 6, Topic A, Lesson 1-4 Mission 6, Topic B, Lesson 5-9 Mission 6, Topic C, Lesson 10-15
2.OA.D.5	Identify arithmetic patterns in an addition or hundreds chart and explain them using properties of operations.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 17 Mission 5, Topic A, Lesson 1 Mission 6, Topic D, Lesson 19
Numbers & Operations in Base 10		
2.NBT.A.1	Know that the three digits of a three-digit number represent amounts of hundreds, tens, and ones (e.g., 706 can be represented in multiple ways as 7 hundreds, 0 tens, and 6 ones; 706 ones; or 70 tens and 6 ones).	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 11-15

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
2nd Grade Standards		**Lessons**
2.NBT.A.2	Recognize, describe, extend, and create patterns when counting by ones, twos, fives, tens, and hundreds and use those patterns to predict the next number in the counting sequence up to 1000 through counting. For example: 111, 113, 115, ...; 82, 84, 86, ...; 370, 380, 390....; 100, 200, 300,...; etc.	Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 12-15 Mission 3, Topic G, Lesson 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
2.NBT.A.3	Read and write numbers to 1000 using standard form, word form, and expanded form. For example, write 234 as 200 + 30 + 4.	Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9
2.NBT.A.4	Compare two three-digit numbers based on the meanings of the digits in each place and use the symbols >, =, and < to show the relationship.	Mission 3, Topic F, Lesson 16-18
2.NBT.B.5	Fluently add and subtract within 100 using properties of operations, strategies based on place value, and/or the relationship between addition and subtraction.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 2-4 Mission 4, Topic B, Lesson 6-8, 10 Mission 4, Topic C, Lesson 11-13 Mission 7, Topic B, Lesson 7-8, 11-13
2.NBT.B.6	Add up to four two-digit numbers using properties of operations and strategies based on place value.	Mission 4, Topic D, Lesson 22
2.NBT.B.7	Add and subtract within 1000 using concrete models, drawings, strategies based on place value, properties of operations, and/or the relationship between addition and subtraction to explain the reasoning used.	Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 14-15 Mission 4, Topic D, Lesson 17-21 Mission 4, Topic E, Lesson 23-28 Mission 4, Topic F, Lesson 29-30 Mission 5, Topic A, Lesson 1-7 Mission 5, Topic B, Lesson 8-12 Mission 5, Topic C, Lesson 13-18 Mission 5, Topic D, Lesson 19-20
2.NBT.B.8	Mentally add or subtract 10 or 100 to/from any given number within 1000.	Mission 3, Topic G, Lesson 19-20 Mission 4, Topic D, Lesson 17 Mission 5, Topic A, Lesson 1-2
Measurement & Data		

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
2nd Grade Standards		**Lessons**
2.MD.A.1	Measure the length of an object in whole number units by selecting and using appropriate tools such as rulers, yardsticks, meter sticks, and measuring tapes.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4-5 Mission 2, Topic C, Lesson 6 Mission 7, Topic C, Lesson 14-17
2.MD.A.2	Measure the length of an object using two different whole number units of measure and describe how the two measurements relate to the size of the unit chosen.	Mission 2, Topic C, Lesson 7 Mission 7, Topic D, Lesson 18
2.MD.A.3	Estimate lengths using whole number units of inches, feet, yards, centimeters, and meters.	Mission 2, Topic B, Lesson 5 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 17
2.MD.A.4	Measure, using whole number lengths, to determine how much longer one object is than another and express the difference in terms of a standard unit of length.	Mission 2, Topic C, Lessons 6, 7 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 19
2.MD.B.5	Add and subtract within 100 to solve contextual problems, with the unknown in any position, involving lengths that are given in the same units by using drawings and equations with a symbol for the unknown to represent the problem.	Mission 2, Topic D, Lessons 8-10 Mission 7, Topic E, Lessons 20, 22
2.MD.B.6	Represent whole numbers as lengths from 0 on a number line and know that the points corresponding to the numbers on the number line are equally spaced. Use a number line to represent whole number sums and differences of lengths within 100.	Mission 2, Topic D, Lesson 8 Mission 7, Topic A, Lesson 3 Mission 7, Topic E, Lessons 21-22 Mission 7, Topic F, Lessons 24
2.MD.C.7	Tell and write time in quarter hours and to the nearest five minutes (in a.m. and p.m.) using analog and digital clocks.	Mission 8, Topic D, Lessons 13-15
2.MD.C.8	Solve contextual problems involving amounts less than one dollar including quarters, dimes, nickels, and pennies using the ¢ symbol appropriately. Solve contextual problems involving whole number dollar amounts up to $100 using the $ symbol appropriately.	Mission 3, Topic D, Lesson 10 Mission 7, Topic B, Lesson 6-10, 12-13
2.MD.D.9	Given a set of data, create a line plot, where the horizontal scale is marked off in whole-number units.	Mission 7, Topic F, Lessons 23-26

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
2nd Grade Standards		Lessons
2.MD.D.10	Draw a pictograph (with a key of values of 1, 2, 5, or 10) and a bar graph (with intervals of one) to represent a data set with up to four categories. Solve addition and subtraction problems related to the data in a graph.	Mission 7, Topic A, Lessons 1-5
Geometry		
2.G.A.1	Identify triangles, quadrilaterals, pentagons, and hexagons. Draw two- dimensional shapes having specified attributes (as determined directly or visually, not by measuring), such as a given number of angles/vertices or a given number of sides of equal length.	Mission 8, Topic A, Lesson 1-6
2.G.A.2	Partition a rectangle into rows and columns of same-sized squares and find the total number of squares.	Mission 6, Topic C, Lesson 14-15
2.G.A.3	Partition circles and rectangles into two, three, and four equal shares. Describe the shares using the words halves, thirds, fourths, half of, a third of, and a fourth of, and describe the whole as two halves, three thirds, four fourths. Recognize that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B, Lesson 7-8 Mission 8, Topic C, Lesson 9-13 Mission 8, Topic D, Lesson 13

3rd Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
Operations & Algebraic Thinking		
3.OA.A.1	Interpret the factors and products in whole number multiplication equations (e.g., 4 x 7 is 4 groups of 7 objects with a total of 28 objects or 4 strings measuring 7 inches each with a total length of 28 inches).	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15, 17 Mission 3, Topic D, Lesson 12 Mission 3, Topic E, Lesson 16
3.OA.A.2	Interpret the dividend, divisor, and quotient in whole number division equations (e.g., 28 ÷ 7 can be interpreted as 28 objects divided into 7 equal groups with 4 objects in each group or 28 objects divided so there are 7 objects in each of the 4 equal groups).	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic D, Lesson 12-13 Mission 1, Topic E, Lesson 17
3.OA.A.3	Multiply and divide within 100 to solve contextual problems, with the unknown in any positions, in situations involving equal groups, arrays/area, and measurement quantities using strategies based on place value, the properties of operations, and the relationship between multiplication and division (e.g., contexts including computations such as 3 x � = 24, 6 x 16 = �, � ÷ 8 = 3, or 96 ÷ 6 = �).	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 8-9 Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 8 Mission 3, Topic D, Lesson 15
3.OA.A.4	Determine the unknown whole number in a multiplication or division equation relating three whole numbers within 100. For example, determine the unknown number that makes the equation true in each of the equations: 8 x � = 48, 5 = � ÷ 3, 6 x 6 =�.	Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 8 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 17 Mission 3, Topic A, Lesson 3 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16
3.OA.B.5	Apply properties of operations as strategies to multiply and divide. (Students need not use formal terms for these properties.)	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15-16 Mission 1, Topic F, Lesson 18-19 Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 9-10 Mission 3, Topic D, Lesson 12-14

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 20
3.OA.B.6	Understand division as an unknown-factor problem. For example, find 32 ÷ 8 by finding the number that makes 32 when multiplied by 8.	Mission 1, Topic B, Lesson 6 Mission 1, Topic D, Lesson 11 Mission 1, Topic E, Lesson 17 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16
3.OA.C.7	Fluently multiply and divide within 100, using strategies such as the properties of operations or the relationship between multiplication and division (e.g., knowing that 8 x 5 = 40, one knows 40 ÷ 5 = 8). By the end of 3rd grade, know all products of two one-digit numbers and related division facts.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14 Mission 1, Topic E, Lesson 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.OA.D.8	Solve two-step contextual problems using the four operations. Represent these problems using equations with a letter standing for the unknown quantity. Assess the reasonableness of answers using mental computation and estimation strategies including rounding.	Mission 1, Topic F, Lesson 20-21 Mission 3, Topic E, Lesson 18 Mission 3, Topic F, Lesson 21 Mission 7, Topic A, Lesson 1-3
3.OA.D.9	Identify patterns in a multiplication chart and explain them using properties of operations.	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
Numbers & Operations in Base 10		
3.NBT.A.1	Round whole numbers to the nearest 10 or 100 using understanding of place value and use a number line to explain how the number was rounded.	Mission 2, Topic C, Lesson 12-14 Mission 2, Topic D, Lesson 17 Mission 2, Topic E, Lesson 20-21
3.NBT.A.2	Fluently add and subtract within 1000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 2, Topic A, Lesson 4-5 Mission 2, Topic B, Lesson 10-11 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
3rd Grade Standards		**Lessons**
3.NBT.A.3	Multiply one-digit whole numbers by multiples of 10 in the range 10–90 (e.g., 9 x 80, 5 x 60) using strategies based on place value and properties of operations.	Mission 3, Topic F, Lesson 19-20
3.NBT.A.4	Read and write multi-digit whole numbers (less than or equal to 100,000) using standard form, word form, and expanded form (e.g., 23,456 can be written as 20,000 + 3,000 + 400 + 50 + 6).	Mission 2, Topic C, Lesson 12-14 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.NF.A.1	Understand a unit fraction, 1/b, as the quantity formed by 1 part when a whole is partitioned into b equal parts; understand a non-unit fraction, n/b, as the quantity formed by n parts of size 1/b. For example, 3/4 represents a quantity formed by 3 parts of size 1/4.	Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10, 12-13 Mission 5, Topic E, Lesson 20, 22, 24-25, 27 Mission 5, Topic F, Lesson 28-29
3.NF.A.2	Understand a fraction as a number on the number line. Represent fractions on a number line.	Mission 5, Topic C 10-11, 13, 16-29
3.NF.A.3	Explain equivalence of fractions and compare fractions by reasoning about their size.	Mission 5, Topic C, Lessons 10-11, 13 Mission 5, Topic D, Lessons 16-19 Mission 5, Topic E, Lessons 20-27 Mission 5, Topic F, Lessons 28-29
Measurement & Data		
3.MD.A.1	Solve contextual problems in time and money.	Mission 2, Topic A, Lesson 1-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21
3.MD.A.2	Measure the mass of objects and liquid volume using standard units of grams (g), kilograms (kg), milliliters (ml), and liters (l). Estimate the mass of objects and liquid volume using benchmarks. For example, a large paper clip is about one gram, so a box of about 100 large clips is about 100 grams.	Mission 3, Topic B, Lesson 6 Mission 2, Topic B, Lesson 7-11 Mission 2, Topic C, Lesson 12 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.MD.B.3	Draw a pictograph and a scaled bar graph to represent a data set with several categories. Solve one- and two-step "how many more" and "how many less" problems using information presented in graphs.	Mission 6, Topic A, Lessons 1-4

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
3rd Grade Standards		**Lessons**
3.MD.B.4	Generate measurement data by measuring lengths using rulers marked with halves and fourths of an inch. Show the data by making a line plot, where the horizontal scale is marked off in appropriate units: whole numbers, halves, or quarters.	Mission 6, Topic B, Lesson 5-9 Mission 7, Topic D, Lesson 19 Mission 7, Topic D, Lesson 22 Mission 7, Topic E, Lesson 26
3.MD.C.5	Recognize that plane figures have an area and understand concepts of area measurement.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-7 Mission 4, Topic D, Lesson 13
3.MD.C.6	Measure areas by counting unit squares (square centimeters, square meters, square inches, square feet, and improvised units).	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-7 Mission 4, Topic D, Lesson 13
3.MD.C.7	Relate area of rectangles to the operations of multiplication and addition.	Mission 4, Topic A, Lesson 4 Mission 4, Topic B, Lesson 5-8 Mission 4, Topic C, Lesson 9-11 Mission 4, Topic D, Lesson 12-16 Mission 7, Topic E, Lesson 28-30
3.MD.D.8	Solve real-world and mathematical problems involving perimeters of polygons, including finding the perimeter given the side lengths, finding an unknown side length, and exploring rectangles with the same perimeter and different areas or with the same area and different perimeters.	Mission 7, Topic C, Lesson 10 Mission 7, Topic C, Lesson 11-17 Mission 7, Topic D, Lesson 18-22 Mission 7, Topic E, Lesson 23-30
Geometry		
3.G.A.1	Understand that shapes in different categories may share attributes and that the shared attributes can define a larger category. Recognize rhombuses, rectangles, and squares as examples of quadrilaterals and recognize examples of quadrilaterals that do not belong to any of these subcategories.	Mission 7, Topic B, Lesson 4-9
3.G.A.2	Partition shapes into parts with equal areas. Recognize that equal shares of identical wholes need not have the same shape. Express the area of each part as a unit fraction of the whole.	Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 24-25, 27 Mission 5, Topic F, Lesson 28-29

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
3rd Grade Standards		Lessons
3.G.A.3	Determine if a figure is a polygon.	Mission 7, Topic B, Lesson 5

4th Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
4th Grade Standards		Lessons
Operations & Algebraic Thinking		
4.OA.A.1	Interpret a multiplication equation as a comparison (e.g., interpret 35 = 5 x 7 as a statement that 35 is 5 times as many as 7 and 7 times as much as 5). Represent verbal/written statements of multiplicative comparisons as multiplication equations.	Mission 1, Topic A, Lesson 1
4.OA.A.2	Multiply or divide to solve contextual problems involving multiplicative comparison, and distinguish multiplicative comparison from additive comparison.	Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9-11 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic G, Lesson 26 Mission 5, Topic G, Lesson 39-40 Mission 7, Topic A, Lesson 4 Mission 7, Topic C, Lesson 14
4.OA.A.3	Solve multi-step contextual problems (posed with whole numbers and having whole-number answers using the four operations) including problems in which remainders must be interpreted. Represent these problems using equations with a letter standing for the unknown quantity.	Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 14, 19 Mission 3, Topic G, Lesson 31-32
4.OA.B.4	Find factor pairs for whole numbers in the range 1–100 using models. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number is prime or composite and whether the given number is a multiple of a given one-digit number.	Mission 3, Topic F, Lesson 22-25
4.OA.C.5	Generate a number or shape pattern that follows a given rule. Identify apparent features of the pattern that were not explicit in the rule itself.	Mission 5, Topic H, Lesson 41
Numbers & Operations in Base 10		
4.NBT.A.1	Recognize that in a multi-digit whole number (less than or equal to 1,000,000), a digit in one place	Mission 1, Topic A, Lesson 2-3

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
	represents 10 times as much as it represents in the place to its right. For example, recognize that 7 in 700 is 10 times bigger than the 7 in 70 because 700 ÷ 70 = 10 and 70 x 10 = 700.	
4.NBT.A.2	Read and write multi-digit whole numbers (less than or equal to 1,000,000) using standard form, word form, and expanded notation (e.g. the expanded notation of 4256 is written as (4 x 1000) + (2 x 100) + (5 x 10) + (6 x 1)). Compare two multi-digit numbers based on meanings of the digits in each place and use the symbols >, =, and < to show the relationship.	Mission 1, Topic A, Lesson 3 Mission 1, Topic B, Lesson 4-6
4.NBT.A.3	Round multi-digit whole numbers to any place (up to and including the hundred-thousand place) using understanding of place value and use a number line to explain how the number was rounded.	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16-17 Mission 1, Topic F, Lesson 19
4.NBT.B.4	Fluently add and subtract within 1,000,000 using efficient strategies and algorithms.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18-19
4.NBT.B.5	Multiply a whole number of up to four digits by a one-digit whole number and multiply two two-digit numbers, using strategies based on place value and the properties of operations. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-11 Mission 3, Topic H, Lesson 34-38
4.NBT.B.6	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic E, Lesson 15-21 Mission 3, Topic G, Lesson 26-33
Numbers & Operations - Fractions		
4.NF.A.1	Explain why a fraction a/b is equivalent to a fraction (a x n)/(b x n) or (a ÷ n)/(b ÷ n) using visual fraction models, with attention to how the number and size of the parts differ even though the two fractions	Mission 5, Topic B, Lesson 7-11 Mission 5, Topic C, Lesson 14-15 Mission 5, Topic D, Lesson 21 Mission 5, Topic E, Lesson 27

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
	themselves are the same size. Use this principle to recognize and generate equivalent fractions. For example, 3/4 = (3 x 2)/(4 x 2) = 6/8.	Mission 6, Topic B, Lesson 5 Mission 6, Topic D, Lesson 12
4.NF.A.2	Compare two fractions with different numerators and different denominators by creating common denominators or common numerators or by comparing to a benchmark such as 0 or 1/2 or 1. Recognize that comparisons are valid only when the two fractions refer to the same whole. Use the symbols >, =, or < to show the relationship and justify the conclusions.	Mission 5, Topic C, Lesson 12-15 Mission 5, Topic E, Lesson 26-27
4.NF.B.3	Understand a fraction a/b with a > 1 as a sum of fractions 1/b. For example, 4/5 = 1/5 + 1/5 + 1/5 + 1/5.	Mission 5, Topic A, Lesson 1-6 Mission 5, Topic D, Lesson 16-21 Mission 5, Topic E, Lesson 22-25 Mission 5, Topic F, Lesson 29-34
4.NF.B.4	Apply and extend understanding of multiplication as repeated addition to multiply a whole number by a fraction.	Mission 5, Topic A, Lesson 3, 5-6 Mission 5, Topic E, Lesson 23-25 Mission 5, Topic G, Lesson 35-40
4.NF.C.5	Express a fraction with denominator 10 as an equivalent fraction with denominator 100, and use this technique to add two fractions with respective denominators 10 and 100.	Mission 6, Topic B, Lesson 4 Mission 6, Topic D, Lesson 12-14
4.NF.C.6	Read and write decimal notation for fractions with denominators 10 or 100. Locate these decimals on a number line.	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-6 Mission 6, Topic C, Lesson 11 Mission 6, Topic D, Lesson 12-14 Mission 6, Topic E, Lesson 15
4.NF.C.7	Compare two decimals to hundredths by reasoning about their size. Recognize that comparisons are valid only when the two decimals refer to the same whole. Use the symbols >, =, or < to show the relationship and justify the conclusions.	Mission 6, Topic C, Lesson 9-11
Measurement & Data		
4.MD.A.1	Measure and estimate to determine relative sizes of measurement units within a single system of	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
	measurement involving length, liquid volume, and mass/weight of objects using customary and metric units.	Mission 4, Topic A, Lesson 4 Mission 7, Topic A, Lesson 1-3 Mission 7, Topic B, Lesson 6-9 Mission 7, Topic C, Lesson 12-13
4.MD.A.2	Solve one- or two-step real-world problems involving whole number measurements (including length, liquid volume, mass/weight, time, and money) with all four operations within a single system of measurement.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 5 Mission 5, Topic G, Lesson 40 Mission 6, Topic D, Lesson 14 Mission 6, Topic E, Lesson 15-16 Mission 7, Topic A, Lesson 1-5 Mission 7, Topic B, Lesson 6-11 Mission 7, Topic C, Lesson 14
4.MD.A.3	Know and apply the area and perimeter formulas for rectangles in real- world and mathematical contexts. For example, find the width of a rectangular room given the area of the flooring and the length, by viewing the area formula as a multiplication equation with an unknown factor.	Mission 3, Topic A, Lesson 1-3 Mission 7, Topic D, Lesson 15-16
4.MD.B.4	Make a line plot to display a data set of measurements in fractions of the same unit (1/2 or 1/4 or 1/8). Use operations on fractions for this grade to solve problems involving information presented in line plots. For example, from a line plot find and interpret the difference in length between the longest and shortest specimens in an insect collection.	Mission 5, Topic E, Lesson 28 Mission 5, Topic G, Lesson 40
4.MD.C.5	Recognize angles as geometric shapes that are formed wherever two rays share a common endpoint; and understand concepts of angle measurement.	Mission 4, Topic A, Lesson 1 Mission 4, Topic B, Lesson 5-7
4.MD.C.6	Measure angles in whole-number degrees using a protractor. Sketch angles of specified measure.	Mission 4, Topic B, Lesson 5-8
4.MD.C.7	Recognize angle measure as additive. When an angle is decomposed into non-overlapping parts, the angle measure of the whole is the sum of the angle measures of the parts. Solve addition and subtraction problems to find unknown angles on a diagram in real-world and mathematical problems. (e.g., by using an equation with a symbol for the unknown angle measure).	Mission 4, Topic C, Lesson 9-11

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
4th Grade Standards		**Lessons**
Geometry		
4.G.A.1	Draw points, lines, line segments, rays, angles (right, acute, obtuse, straight, reflex), and perpendicular and parallel lines. Identify these in two- dimensional figures.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic D, Lesson 13-15
4.G.A.2	Classify two-dimensional figures based on the presence or absence of parallel or perpendicular lines or the presence or absence of angles of a specified size. Classify triangles based on the measure of the angles as right, acute, or obtuse.	Mission 4, Topic D, Lesson 13-15
4.G.A.3	Recognize and draw lines of symmetry for two-dimensional figures.	Mission 4, Topic D, Lesson 12

5th Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
Operations & Algebraic Thinking		
5.OA.A.1	Use parentheses and/or brackets in numerical expressions involving whole numbers and evaluate expressions having these symbols using the conventional order by applying the Order of Operations.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 3-5 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.OA.A.2	Write numerical expressions that record calculations with numbers and interpret numerical expressions without evaluating them. For example, express the calculation "add 8 and 7, then multiply by 2" as 2 x (8 + 7). Recognize that 3 x (18,932 + 921) is three times as large as 18,932 + 921, without having to calculate the indicated sum or product.	Mission 2, Topic B, Lesson 3-4 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.OA.B.3	Generate two numerical patterns using two given rules. For example, given the rule "Add 3" and the starting number 0, and given the rule "Add 6" and the starting number 0, generate terms in the resulting sequences.	Mission 6, Topic B, Lesson 7
Numbers & Operations in Base 10		
5.NBT.A.1	Recognize that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and 1/10 of what it represents in the place to its left.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 2 Mission 2, Topic E, Lesson 16
5.NBT.A.2	Explain patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of 10. Use whole-number exponents to denote powers of 10.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 1 Mission 2, Topic C, Lesson 10 Mission 2, Topic C, Lesson 11-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24
5.NBT.A.3	Read and write decimals to thousandths using standard form, word form, and expanded notation (e.g., the expanded notation of 347.392 is written as (3	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
5th Grade Standards		Lessons
	x 100) + (4 x 10) + (7 x 1) + (3 x (1/10)) + (9 x (1/100)) + (2 x (1/1000))). Compare two decimals to thousandths based on meanings of the digits in each place and use the symbols >, =, and < to show the relationship.	Mission 1, Topic F, Lesson 13-15
5.NBT.A.4	Round decimals to the nearest hundredth, tenth, or whole number using understanding of place value, and use a number line to explain how the number was rounded.	Mission 1, Topic C, Lesson 7-8
5.NBT.B.5	Fluently multiply multi-digit whole numbers (up to three-digit by four-digit factors) using efficient strategies and algorithms.	Mission 2, Topic B, Lesson 5-9 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic F, Lesson 20-23
5.NBT.B.6	Find whole-number quotients and remainders of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 2, Topic E, Lesson 16-18 Mission 2, Topic F, Lesson 19-23 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic G, Lesson 30-31
5.NBT.B.7	Add, subtract, multiply, and divide decimals to hundredths, using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between operations. Assess the reasonableness of answers using estimation strategies.	Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-16 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24-27 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic E, Lesson 17-18 Mission 4, Topic G, Lesson 29-31
Numbers & Operations - Fractions		
5.NF.A.1	Add and subtract fractions with unlike denominators (including mixed numbers) by replacing given fractions with equivalent fractions in such a way as to produce an equivalent sum or difference of fractions with like denominators. For example, 2/3 + 5/4 = 8/12 + 15/12 = 23/12 or 3/5 + 7/10 = 6/10 + 7/10 = 13/10.	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 3-6 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 14
5.NF.A.2	Solve contextual problems involving addition and subtraction of fractions referring to the same whole,	Mission 3, Topic B, Lesson 3-7 Mission 3, Topic C, Lesson 8-12

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
5th Grade Standards		**Lessons**
	including cases of unlike denominators. Use benchmark fractions and number sense of fractions to estimate mentally and assess the reasonableness of answers. For example, recognize an incorrect result 2/5 + 1/2 = 3/7, by observing that 3/7 < 1/2.	Mission 3, Topic D, Lesson 13-15 Mission 4, Topic D, Lesson 10-12
5.NF.B.3	Interpret a fraction as division of the numerator by the denominator (a/b = a ÷ b). For example, 3/4 = 3 ÷ 4 so when 3 wholes are shared equally among 4 people, each person has a share of size 3/4. Solve contextual problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers by using visual fraction models or equations to represent the problem.	Mission 4, Topic B, Lesson 2-5
5.NF.B.4	Apply and extend previous understandings of multiplication to multiply a fraction by a whole number or a fraction by a fraction.	Mission 4, Topic C, Lesson 6-9 Mission 4, Topic E, Lesson 13-18 Mission 5, Topic C, Lesson 10-15
5.NF.B.5	Interpret multiplication as scaling (resizing).	Mission 4, Topic F, Lesson 21-23 Mission 4, Topic H, Lesson 32
5.NF.B.6	Solve real-world problems involving multiplication of fractions and mixed numbers by using visual fraction models or equations to represent the problem.	Mission 4, Topic C, Lesson 6-7 Mission 4, Topic D, Lesson 10-12 Mission 4, Topic E, Lesson 13-18 Mission 4, Topic F, Lesson 22-24 Mission 5, Topic C, Lesson 12-15
5.NF.B.7	Apply and extend previous understandings of division to divide unit fractions by whole numbers and whole numbers by unit fractions.	Mission 4, Topic G, Lesson 25-33
Measurement & Data		
5.MD.A.1	Convert customary and metric measurement units within a single system by expressing measurements of a larger unit in terms of a smaller unit. Use these conversions to solve multi-step real-world problems involving distances, intervals of time, liquid volumes, masses of objects, and money (including problems involving simple fractions or decimals). For example, 3.6 liters and 4.1 liters can be combined as 7.7 liters or 7700 milliliters.	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lesson 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lesson 19-20 Mission 5, Topic B, Lesson 5

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
5th Grade Standards		**Lessons**
5.MD.B.2	Make a line plot to display a data set of measurements in fractions of a unit (1/2, 1/4, 1/8). Use operations on fractions for this grade to solve problems involving information presented in line plots.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 10
5.MD.C.3	Recognize volume as an attribute of solid figures and understand concepts of volume measurement.	Mission 5, Topic A, Lesson 1-3 Mission 5, Topic B, Lesson 5
5.MD.C.4	Measure volume by counting unit cubes, using cubic centimeters, cubic inches, cubic feet, and improvised units.	Mission 5, Topic A, Lesson 1-3
5.MD.C.5	Relate volume to the operations of multiplication and addition and solve real-world and mathematical problems involving volume of right rectangular prisms.	Mission 5, Topic B, Lesson 4-9
Geometry		
5.G.A.1	Graph ordered pairs and label points using the first quadrant of the coordinate plane. Understand in the ordered pair that the first number indicates the horizontal distance traveled along the x-axis from the origin and the second number indicates the vertical distance traveled along the y-axis, with the convention that the names of the two axes and the coordinates correspond (e.g., x-axis and x- coordinate, y-axis and y-coordinate).	Mission 6, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 7-12 Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 18, 20
5.G.A.2	Represent real-world and mathematical problems by graphing points in the first quadrant of the coordinate plane and interpret coordinate values of points in the context of the situation.	Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 18-20
5.G.B.3	Classify two-dimensional figures in a hierarchy based on properties. Understand that attributes belonging to a category of two-dimensional figures also belong to all subcategories of that category. For example, all rectangles have four right angles and squares are rectangles, so all squares have four right angles.	Mission 5, Topic D, Lesson 16-21

6th Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
Ratios & Proportional Relationships		
6.RP.A.1	Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities. Make a distinction between ratios and fractions.	Mission 2, Topic A, Lesson 1-2 Mission 2, Topic B, Lesson 3-5 Mission 9, Topic A, Lesson 4
6.RP.A.2	Understand the concept of a unit rate a/b associated with a ratio a:b with $b \neq 0$. Use rate language in the context of a ratio relationship.	Mission 2, Topic C, Lesson 10 Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 5-7 Mission 9, Topic A, Lesson 6
6.RP.A.3	Use ratio and rate reasoning to solve real-world and mathematical problems (e.g., by reasoning about tables of equivalent ratios, tape diagrams, double number line diagrams, or equations).	Mission 2, Topic C, Lesson 6-7, 10 Mission 2, Topic D, Lesson 12-14 Mission 2, Topic E, Lesson 15-16 Mission 2, Topic F, Lesson 17 Mission 3, Topic C, Lesson 6-9 Mission 3, Topic D, Lesson 15 Mission 9, Topic A, Lesson 4-6 Mission 2, Topic C, Lesson 8-9 Mission 2, Topic D, Lesson 11 Mission 3, Topic C, Lesson 5 Mission 6, Topic D, Lesson 16-17 Mission 3, Topic D, Lesson 10-14, 16 Mission 6, Topic B, Lesson 7
The Number System		
6.NS.A.1	Interpret and compute quotients of fractions, and solve real-world and mathematical problems involving division of fractions by fractions (e.g., connecting visual fraction models and equations to represent the problem is suggested).	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4 Mission 4, Topic B, Lesson 5-9 Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
6.NS.B.2	Fluently divide multi-digit numbers using a standard algorithm.	Mission 5, Topic D, Lessons 9-11

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
6th Grade Standards		**Lessons**
6.NS.B.3	Fluently add, subtract, multiply, and divide multi-digit decimals using a standard algorithm and making connections to previous conceptual work with each operation.	Mission 5, Topic B, Lesson 2-4 Mission 5, Topic C, Lesson 7-8 Mission 5, Topic D, Lesson 12-13 Mission 5, Topic E, Lesson 14-15 Mission 6, Topic A, Lesson 4 Mission 8, Topic C, Lesson 12 Mission 9, Topic A, Lesson 6
6.NS.B.4	Find the greatest common factor of two whole numbers less than or equal to 100 and the least common multiple of two whole numbers less than or equal to 12. Use the distributive property to express a sum of two whole numbers 1–100 with a common factor as a multiple of a sum of two whole numbers with no common factor.	Mission 7, Topic D, Lesson 16-18
6.NS.C.5	Understand that positive and negative numbers are used together to describe quantities having opposite directions or values (e.g., temperature above/below zero, elevation above/below sea level, credits/debits, positive/negative electric charge); use positive and negative numbers to represent quantities in real-world contexts, explaining the meaning of 0 in each situation as well as describing situations in which opposite quantities can combine to make 0.	Mission 7, Topic A, Lesson 1, 5
6.NS.C.6	Understand a rational number as a point on the number line. Extend number line diagrams and coordinate axes familiar from previous grades to represent points on the line and in the plane with negative number coordinates.	Mission 7, Topic A, Lesson 1-2 Mission 7, Topic A, Lesson 4, 7 Mission 7, Topic C, Lesson 14 Mission 7, Topic C, Lesson 11-13, 15
6.NS.C.7	Understand ordering and absolute value of rational numbers.	Mission 7, Topic A, Lesson 3-4 Mission 7, Topic A, Lesson 6-7 Mission 7, Topic B, Lesson 8-9 Mission 7, Topic C, Lesson 13
6.NS.C.8	Solve real-world and mathematical problems by graphing points in all four quadrants of the coordinate plane. Include use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	Mission 7, Topic C, Lesson 11, 13-15
Expressions & Equations		

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
6.EE.A.1	Write and evaluate numerical expressions involving whole-number exponents.	Mission 1, Topic F, Lesson 17-18 Mission 6, Topic C, Lesson 12-15
6.EE.A.2	Write, read, and evaluate expressions in which variables stand for numbers.	Mission 6, Topic B, Lesson 10-11 Mission 1, Topic B, Lesson 5 Mission 1, Topic C, Lesson 9 Mission 1, Topic F, Lesson 18 Mission 6, Topic A, Lesson 2 Mission 6, Topic B, Lesson 6 Mission 6, Topic C, Lesson 14-15 Mission 7, Topic D, Lesson 16
6.EE.A.3	Apply the properties of operations (including, but not limited to, commutative, associative, and distributive properties) to generate equivalent expressions.	Mission 6, Topic B, Lesson 8 Mission 6, Topic B, Lesson 10-11
6.EE.A.4	Identify when expressions are equivalent (i.e., when the expressions name the same number regardless of which value is substituted into them).	Mission 5, Topic D, Lesson 13 Mission 6, Topic B, Lesson 8 Mission 6, Topic B, Lesson 10-11
6.EE.B.5	Understand that a solution to an equation or inequality is the value(s) that makes that statement true. Use substitution to determine whether a given number in a specified set makes an equation or inequality true.	Mission 6, Topic A, Lesson 2-5 Mission 6, Topic B, Lesson 8 Mission 6, Topic C, Lesson 15 Mission 7, Topic B, Lesson 9-10
6.EE.B.6	Use variables to represent numbers and write expressions when solving real-world and mathematical problems; understand that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set.	Mission 6, Topic A, Lesson 1, 3-5 Mission 6, Topic B, Lesson 6-8 Mission 7, Topic B, Lesson 10
6.EE.B.7	Solve real-world and mathematical problems by writing and solving one- step equations of the form $x + p = q$, $px = q$, $x - p = q$, and $x/p = q$ for cases in which p, q, and x are all nonnegative rational numbers and $p \neq 0$. (Complex fractions are not an expectation at this grade level.)	Mission 6, Topic A, Lesson 3-5 Mission 6, Topic B, Lesson 7
6.EE.B.8	Interpret and write an inequality of the form $x > c$, $x < c$, $x \leq c$, or $x \geq c$ which represents a condition or constraint in a real-world or mathematical problem. Recognize that inequalities have infinitely many	Mission 7, Topic B, Lesson 8-10

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
6th Grade Standards		Lessons
	solutions; represent solutions of inequalities on number line diagrams.	
6.EE.C.9	Use variables to represent two quantities in a real-world problem that change in relationship to one another.	Mission 6, Topic D, Lesson 16-18
Geometry		
6.G.A.1	Find the area of right triangles, other triangles, special quadrilaterals, and polygons by composing into rectangles or decomposing into triangles and other shapes; know and apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic A, Lesson 2-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 11 Mission 1, Topic G, Lesson 19 Mission 4, Topic D, Lesson 14
6.G.A.2	Find the volume of a right rectangular prism with fractional edge lengths by packing it with unit cubes of the appropriate unit fraction edge lengths, and show that the volume is the same as would be found by multiplying the edge lengths of the prism. Apply the formulas V = lwh and V = Bh where B is the area of the base to find volumes of right rectangular prisms with fractional edge lengths in the context of solving real-world and mathematical problems.	Mission 1, Topic E, Lesson 15 Mission 4, Topic D, Lesson 14-15 Mission 4, Topic E, Lesson 17
6.G.A.3	Draw polygons in the coordinate plane given coordinates for the vertices; use coordinates to find the length of a side that joins two vertices (vertical or horizontal segments only). Apply these techniques in the context of solving real-world and mathematical problems.	Mission 7, Topic C, Lesson 15
6.G.A.4	Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic E, Lesson 12-16 Mission 1, Topic F, Lesson 18 Mission 1, Topic G, Lesson 19
Statistics & Probability		

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
6th Grade Standards		**Lessons**
6.SP.A.1	Recognize a statistical question as one that anticipates variability in the data related to the question and accounts for it in the answers.	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3, 6-7 Mission 8, Topic D, Lesson 17
6.SP.A.2	Understand that a set of data collected to answer a statistical question has a distribution which can be described by its measures of center (mean, median, mode), measures of variation (range only), and overall shape.	Mission 8, Topic B, Lesson 4-5, 7-8 Mission 8, Topic C, Lesson 11 Mission 8, Topic E, Lesson 18
6.SP.A.3	Recognize that a measure of center (mean, median, mode) for a numerical data set summarizes all of its values with a single number, while a measure of variation describes how its values vary with a single number.	Mission 8, Topic B, Lesson 6 Mission 8, Topic C, Lesson 9-11
6.SP.B.4	Display a single set of numerical data using dot plots (line plots), box plots, pie charts and stem plots.	Mission 8, Topic B, Lesson 3-8 Mission 8, Topic D, Lesson 16-17
6.SP.B.5	Summarize numerical data sets in relation to their context.	Mission 8, Topic D, Lesson 17 Mission 8, Topic B, Lesson 3-4 Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3 Mission 8, Topic B, Lesson 5-7 Mission 8, Topic D, Lesson 14 Mission 8, Topic C, Lesson 9 Mission 8, Topic C, Lesson 10-12 Mission 8, Topic D, Lesson 13-16 Mission 8, Topic E, Lesson 18 Mission 8, Topic C, Lesson 12 Mission 8, Topic D, Lesson 14-16 Mission 8, Topic E, Lesson 18

7th Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
Ratios & Proportional Relationships		
7.RP.A.1	Compute unit rates associated with ratios of fractions, including ratios of lengths, areas, and other quantities measured in like or different units.	Mission 2, Topic C, Lesson 8 Mission 4, Topic A, Lesson 2 Mission 4, Topic A, Lesson 3 Mission 9, Topic B, Lesson 5
7.RP.A.2	Recognize and represent proportional relationships between quantities.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic B, Lesson 4-6 Mission 2, Topic C, Lesson 7-9 Mission 2, Topic D, Lesson 10-13 Mission 2, Topic E, Lesson 14-15 Mission 3, Topic A, Lesson 1, 3, 5 Mission 4, Topic A, Lesson 3-5 Mission 5, Topic C, Lesson 9, 12 Mission 5, Topic D, Lesson 14 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 5 Mission 3, Topic B, Lesson 7
7.RP.A.3	Use proportional relationships to solve multi-step ratio and percent problems.	Mission 3, Topic A, Lesson 5 Mission 4, Topic B, Lesson 6-9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16 Mission 9, Topic A, Lesson 1-4 Mission 9, Topic B, Lesson 6, 8 Mission 9, Topic C, Lesson 13
The Number System		
7.NS.A.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a horizontal or vertical number line diagram.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2-3, 5-7 Mission 6, Topic D, Lesson 18
7.NS.A.2	Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers.	Mission 5, Topic C, Lesson 8-11 Mission 4, Topic A, Lesson 5 Mission 5, Topic A, Lesson 1 Mission 8, Topic D, Lesson 16

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
		Mission 9, Topic A, Lesson 4
7.NS.A.3	Solve real-world and mathematical problems involving the four operations with rational numbers. (Computations with rational numbers extend the rules for manipulating fractions to complex fractions.)	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 12 Mission 5, Topic D, Lesson 13-14 Mission 5, Topic E, Lesson 15-16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6
Expression & Equations		
7.EE.A.1	Apply properties of operations as strategies to add, subtract, factor, and expand linear expressions with rational coefficients.	Mission 6, Topic D, Lesson 18-22 Mission 9, Topic B, Lesson 7
7.EE.A.2	Rewrite and connect equivalent expressions in different forms in a contextual problem to provide multiple ways of interpreting the problem and investigating how the quantities in it are related.	Mission 6, Topic B, Lesson 12
7.EE.B.3	Solve multi-step real-world and mathematical problems posed with positive and negative rational numbers presented in any form (whole numbers, fractions, and decimals).	Mission 3, Topic C, Lesson 11 Mission 5, Topic C, Lesson 12 Mission 5, Topic F, Lesson 17 Mission 6, Topic A, Lesson 2-6 Mission 6, Topic B, Lesson 11-12
7.EE.B.4	Use variables to represent quantities in a real-world and mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities.	Mission 5, Topic C, Lesson 15 Mission 6, Topic A, Lesson 5 Mission 6, Topic B, Lesson 9 Mission 6, Topic B, Lesson 11-12 Mission 6, Topic C, Lesson 13, 15 Mission 7, Topic A, Lesson 5 Mission 9, Topic A, Lesson 3 Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 9, Topic B, Lesson 7 Mission 6, Topic C, Lesson 14, 16-17
Geometry		

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
7th Grade Standards		**Lessons**
7.G.A.1	Solve problems involving scale drawings of congruent and similar geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7-12 Mission 1, Topic C, Lesson 13 Mission 2, Topic A, Lesson 1 Mission 3, Topic B, Lesson 6 Mission 3, Topic C, Lesson 11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 13
7.G.A.2	Draw triangles with given conditions: three angle measures or three side measures. Notice when the conditions determine a unique triangle, more than one triangle, or no triangle.	Mission 3, Topic A, Lesson 2 Mission 7, Topic B, Lesson 6-10 Mission 7, Topic D, Lesson 17
7.G.B.3	Know the formulas for the area and circumference of a circle and use them to solve problems. Explore the relationships between the radius, the circumference, and the area of a circle, and the number π.	Mission 3, Topic A, Lesson 3-5 Mission 3, Topic B, Lesson 7-9 Mission 3, Topic C, Lesson 10-11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 11-12
7.G.B.4	Know and use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle in a figure.	Mission 7, Topic A, Lesson 2-5
7.G.B.5	Solve real-world and mathematical problems involving area of two- dimensional figures composed of triangles, quadrilaterals, and polygons, and volume and surface area of three-dimensional objects composed of cubes and right prisms.	Mission 1, Topic A, Lesson 6 Mission 2, Topic C, Lesson 8 Mission 3, Topic B, Lesson 6 Mission 7, Topic C, Lesson 12-16 Mission 7, Topic D, Lesson 17 Mission 9, Topic A, Lesson 4 Mission 9, Topic B, Lesson 5, 9
Statistics & Probability		
7.SP.A.1	Explore how statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. Understand that random sampling tends to produce representative samples and support valid inferences.	Mission 8, Topic C, Lesson 12-14 Mission 8, Topic D, Lesson 15 Mission 8, Topic E, Lesson 20

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
7th Grade Standards		Lessons
7.SP.A.2	Collect and use data from a random sample to draw inferences about a population with an unknown characteristic of interest. Generate multiple samples (or simulated samples) of the same size to gauge the variation in estimates or predictions.	Mission 8, Topic C, Lesson 13-14 Mission 8, Topic D, Lesson 15-17 Mission 8, Topic E, Lesson 20
7.SP.B.3	Informally compare the measures of center (mean, median, mode) of two numerical data distributions with similar variabilities.	Mission 8, Topic C, Lesson 11 Mission 8, Topic D, Lesson 18
7.SP.B.4	Use measures of center and measures of variability for numerical data from random samples to draw informal comparative inferences about two populations.	Mission 8, Topic D, Lesson 15-16, 18-19 Mission 8, Topic E, Lesson 20 Mission 9, Topic A, Lesson 3
7.SP.C.5	Recognize that the probability of a chance event is a number between 0 and 1 and interpret the likelihood of the event occurring.	Mission 8, Topic A, Lesson 2-6
7.SP.C.6	Calculate theoretical and experimental probability of simple events.	Mission 8, Topic A, Lesson 1, 3-6
7.SP.C.7	Develop a probability model and use it to find experimental or theoretical probabilities of events.	Mission 8, Topic A, Lesson 3-5 Mission 8, Topic C, Lesson 14 Mission 8, Topic A, Lesson 3 Mission 8, Topic E, Lesson 20 Mission 8, Topic A, Lesson 4-6
7.SP.D.8	Summarize a numerical data set in relation to its context.	Mission 8, Topic D, Lesson 15-19

8th Grade

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
The Number System		
8.NS.A.1	Know that real numbers that are not rational are called irrational (e.g., π,√2, etc). Understand informally that every number has a decimal expansion; for rational numbers show that the decimal expansion repeats eventually or terminates, and convert a decimal expansion which repeats eventually or terminates into a rational number.	Mission 8, Topic E, Lesson 14-15
8.NS.A.2	Use rational approximations of irrational numbers to compare the size of irrational numbers by locating them approximately on a number line diagram. Estimate the value of irrational expressions (such as π2).	Mission 8, Topic A, Lesson 1 Mission 8, Topic B, Lesson 4-5 Mission 8, Topic D, Lesson 12-13
Expressions & Equations		
8.EE.A.1	Know and apply the properties of integer exponents to generate equivalent numerical expressions. For example, 3^2 x 3^-5 = 3^-3 = 1/33 = 1/27.	Mission 7, Topic B, Lesson 2-8 Mission 7, Topic C, Lesson 11, 14
8.EE.A.2	Use square root and cube root symbols to represent solutions to equations of the form x^2 = p and x^3 = p, where p is a positive rational number. Evaluate square roots of small perfect squares and cube roots of small perfect cubes.	Mission 8, Topic B, Lesson 2-5 Mission 8, Topic C, Lesson 10 Mission 8, Topic D, Lesson 12-13
8.EE.A.3	Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities and to express how many times as much one is than the other.	Mission 7, Topic C, Lesson 9-12, 14 Mission 7, Topic D, Lesson 16
8.EE.A.4	Using technology, solve real-world problems with numbers expressed in decimal and scientific notation. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities (e.g., use millimeters per year for seafloor spreading).	Mission 7, Topic C, Lesson 10-15 Mission 7, Topic D, Lesson 16

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
8th Grade Standards		**Lessons**
8.EE.B.5	Graph proportional relationships, interpreting the unit rate as the slope of the graph. Compare two different proportional relationships represented in different ways.	Mission 3, Topic A , Lesson 2-4 Mission 3, Topic B, Lesson 6
8.EE.B.6	Use similar triangles to explain why the slope m is the same between any two distinct points on a non-vertical line in the coordinate plane; know and apply the equation y = mx for a line through the origin and the equation y = mx + b for a line intercepting the vertical axis at b.	Mission 2, Topic C, Lesson 10-12 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 10-11 Mission 3, Topic E, Lesson 14
8.EE.C.7	Solve linear equations in one variable.	Mission 4, Topic B, Lesson 3-9
8.EE.C.8	Analyze and solve systems of two linear equations graphically.	Mission 3, Topic C, Lesson 13-14 Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16
8.EE.C.9	By graphing on the coordinate plane or by analyzing a given graph, determine the solution set of a linear inequality in one or two variables.	Mission 3, Topic E, Lesson 15
Functions		
8.F.A.1	Understand that a function is a rule that assigns to each input exactly one output. The graph of a function is the set of ordered pairs consisting of an input and the corresponding output. (Function notation is not required in 8th grade.)	Mission 5, Topic A, Lesson 1-2 Mission 5, Topic B, Lesson 3-5 Mission 5, Topic E, Lesson 17 Mission 9, Topic A, Lesson 1
8.F.A.2	Compare properties of two functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 8
8.F.A.3	Know and interpret the equation y = mx + b as defining a linear function, whose graph is a straight line; give examples of functions that are not linear.	Mission 5, Topic B, Lessons 4, 7 Mission 5, Topic C, Lesson 8 Mission 5, Topic E, Lesson 18
8.F.B.4	Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a	Mission 5, Topic C, Lesson 8-10 Mission 5, Topic D, Lesson 11

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
8th Grade Standards		**Lessons**
	relationship or from two (x, y) values, including reading these from a table or from a graph. Interpret the rate of change and initial value of a linear function in terms of the situation it models and in terms of its graph or a table of values.	
8.F.B.5	Describe qualitatively the functional relationship between two quantities by analyzing a graph (e.g., where the function is increasing or decreasing, linear or nonlinear). Sketch a graph that exhibits the qualitative features of a function that has been described verbally.	Mission 5, Topic B, Lesson 5-6 Mission 5, Topic C, Lesson 10
Geometry		
8.G.A.1	Describe the effect of translations, rotations, reflections, and dilations on two-dimensional figures using coordinates.	Mission 1, Topic A, Lesson 2-4, 6 Mission 1, Topic B, Lesson 7-10 Mission 1, Topic C, Lesson 11, 13 Mission 1, Topic D, Lesson 14 Mission 3, Topic B, Lesson 8 Mission 1, Topic A, Lesson 5-6 Mission 2, Topic A, Lesson 4-5 Mission 2, Topic C, Lesson 12
8.G.A.2	Use informal arguments to establish facts about the angle sum and exterior angle of triangles, about the angles created when parallel lines are cut by a transversal, and the angle-angle criterion for similarity of triangles.	Mission 1, Topic D, Lesson 14-16 Mission 2, Topic B, Lesson 8 Mission 2, Topic D, Lesson 13 Mission 9, Topic A, Lesson 2
8.G.B.3	Explain a model of the Pythagorean Theorem and its converse.	Mission 8, Topic C, Lesson 7, 9
8.G.B.4	Know and apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and mathematical problems in two and three dimensions.	Mission 8, Topic C, Lesson 6-8, 10
8.G.B.5	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.	Mission 8, Topic C, Lesson 11

TENNESSEE STATE MATH STANDARDS		ZEARN MATH
8th Grade Standards		Lessons
8.G.C.6	Apply the formulas for the volumes of cones, cylinders, and spheres to solve real-world and mathematical problems.	Mission 5, Topic D, Lesson 13-16 Mission 5, Topic E, Lesson 17-21 Mission 5, Topic F, Lesson 22
Statistics & Probability		
8.SP.A.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.	Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 3-8
8.SP.A.2	Know that straight lines are widely used to model linear relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line and informally assess the model fit by judging the closeness of the data points to the line.	Mission 6, Topic B, Lesson 4-8
8.SP.A.3	Use the equation of a linear model to solve problems in the context of bivariate measurement data, interpreting the slope and intercepts.	Mission 6, Topic B, Lesson 6, 8
8.SP.B.4	Find probabilities of and represent sample spaces for compound events using organized lists, tables, tree diagrams, and simulation.	Mission 6, Lesson 12, 13